

# OpenWebSearch.eu - Building an Open Index of the Web on HPC Infrastructure

Prof. Dr. Michael Granitzer  
Chair of Data Science, University of Passau  
Coordinator OpenWebSearch.eu

HPCSE 2026 · IT4Innovations · 2026-05-20

# Web Search Today

A critical resource managed as an oligopoly

## Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

Being able to navigate the Web is as fundamental as GPS or the power grid

# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

1.15 Billion

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

Top 0.1%–1% of sites

may account for roughly half of all pages  
heavy-tail intuition from large web graph studies

~ 400 Billion

documents in Google's search index  
2020 figure cited in US v. Google testimony

Sites ≠ domains ≠ pages; use exact counts cautiously

### VELOCITY

+8.6 M / month

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

~ 3.2 / second

same monthly net change, averaged across December

149 ZB (global)

all digital data created / copied worldwide in 2024  
IDC / Statista global datasphere estimate

The web changes constantly; crawling is continuous, not one-off

### VARIETY

150+ languages

observed across websites

W3Techs usage survey

249 CMS Platforms

CMS platforms identified in the HTTP Archive dataset  
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

HTML · PDF · JSON-LD · media · APIs

content types to parse and process

HTML · PDFs · JS apps · schemas · media · APIs

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·  
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

**1.15 Billion**

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

**Top 0.1%–1% of sites**

may account for roughly half of all pages  
heavy-tail intuition from large web graph studies

**~ 400 Billion**

documents in Google's search index  
2020 figure cited in US v. Google testimony

*Sites ≠ domains ≠ pages; use exact counts cautiously*

### VELOCITY

**+8.6 M / month**

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

**~ 3.2 / second**

same monthly net change, averaged across December

**149 ZB (global)**

all digital data created / copied worldwide in 2024  
IDC / Statista global datasphere estimate

*The web changes constantly; crawling is continuous, not one-off*

### VARIETY

**150+ languages**

observed across websites

W3Techs usage survey

**249 CMS Platforms**

CMS platforms identified in the HTTP Archive dataset  
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

**HTML · PDF · JSON-LD · media · APIs**

content types to parse and process

*HTML · PDFs · JS apps · schemas · media · APIs*

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·  
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

High upfront costs to tap the Web as resources  
— especially small innovators and researchers  
are left behind

# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

- 1. Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
- 2. Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
- 3. Added Value Services** Dashboard, tools, data access APIs
- 4. Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners



Funded by **Horizon Europe** (NGI programme)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



# The Open Web Index

# The Open Web Index

**The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.

# The Open Web Index

**The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.



## Federated production

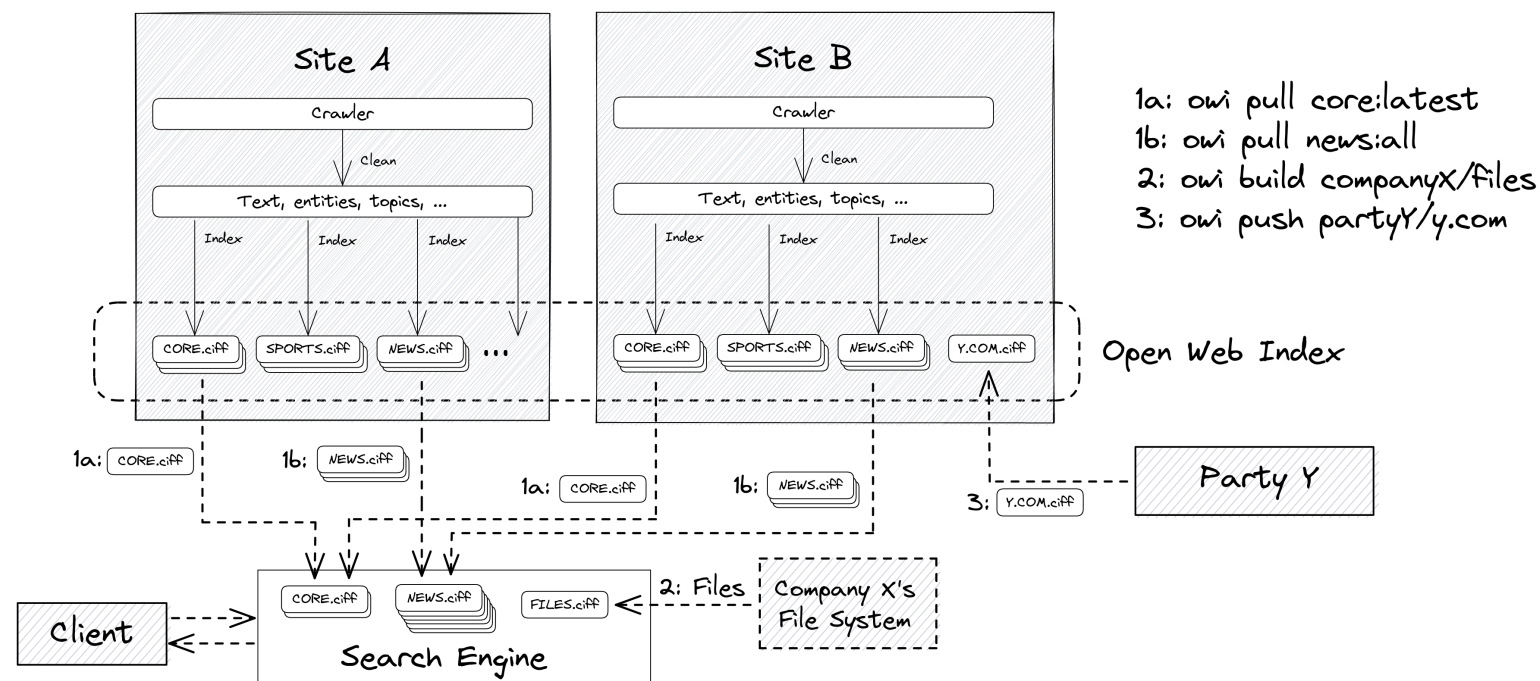
- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

## Three access verbs

- owi pull — download pre-built shards
- owi build — index your own data
- owi push — contribute shards back

# The Open Web Index

**The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.



## Federated production

- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

## Three access verbs

- owi pull — download pre-built shards
- owi build — index your own data
- owi push — contribute shards back

## Three output types per daily run — the building blocks of a shard:

- `index.ciff.gz` — sparse inverted index (CIFF format; pre-configured tokenizer)
- `metadata_*.parquet` — document metadata & enrichments (topics, microdata, locations, language, etc.; extensible)
- Dense embeddings (GPU-enriched slice, Jina V3) for AI-based search and classification (on selected subsets)

# OWI at a Glance — Statistics (Feb 2026)

## Crawling

|                 |                           |
|-----------------|---------------------------|
|                 |                           |
| Pages crawled   | ~39 B total, ~12 B unique |
| Unique hosts    | ~500 M (extrapolated)     |
| Crawling effort | 1,927 machine-days        |
| Avg. throughput | ~20 M URLs / day          |

## Index

|                   |                            |
|-------------------|----------------------------|
|                   |                            |
| Documents indexed | ~10.7 B                    |
| Unique domains    | ~75 M                      |
| Public datasets   | 511 main / 1,511 total     |
| Index size        | 45 TB (main) / 50 TB total |
| Embeddings        | 119 M docs · 884 GB        |

**Top languages:** English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)

**Multimedia Objects:** Roughly 2-3 quality images and 0.5 Videos per Web-Document

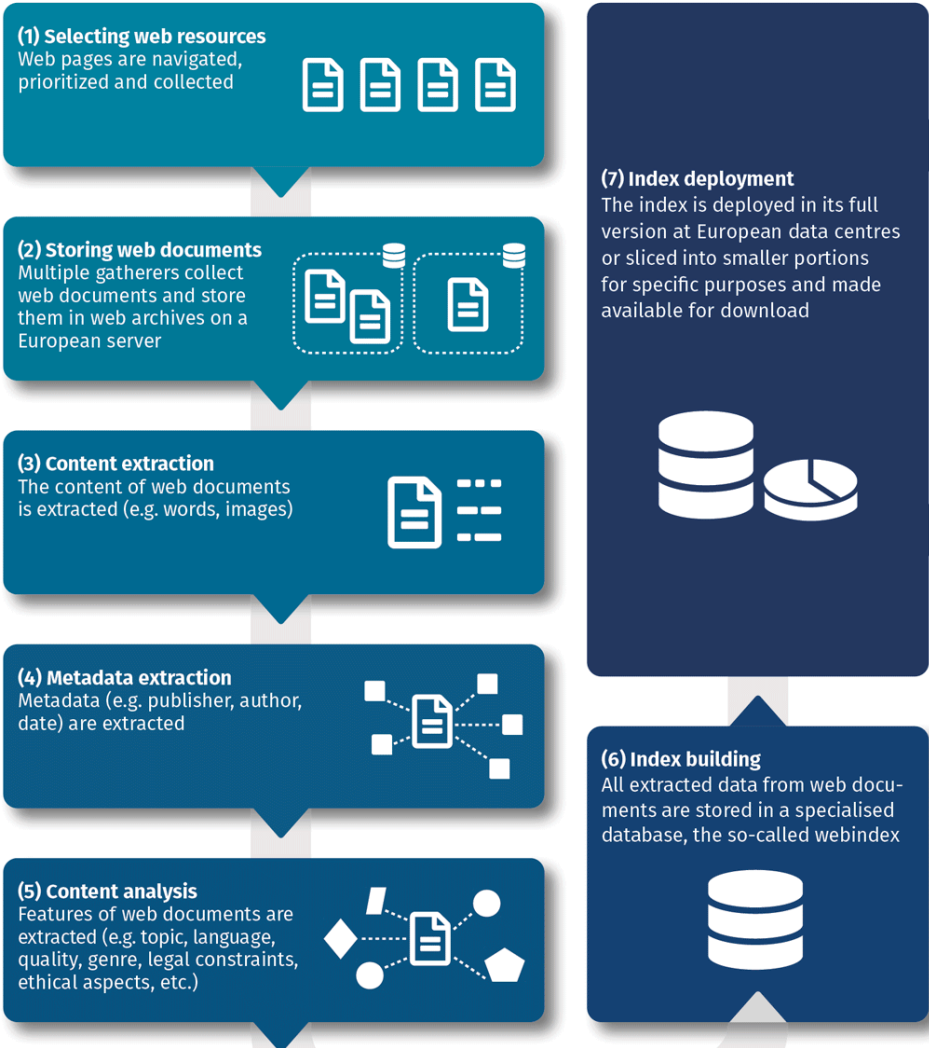
**Comparison:** To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

# Building the Open Web Index on HPC Infrastructure

# Conceptual Workflow

## Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.



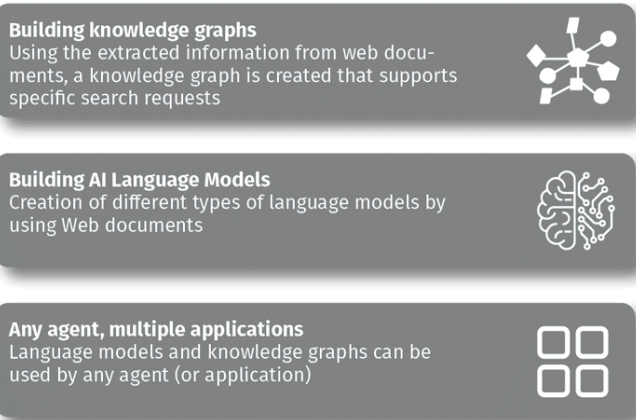
## Search Applications

A user search request will be answered by a search application that makes use of the open web index.



## Data Products

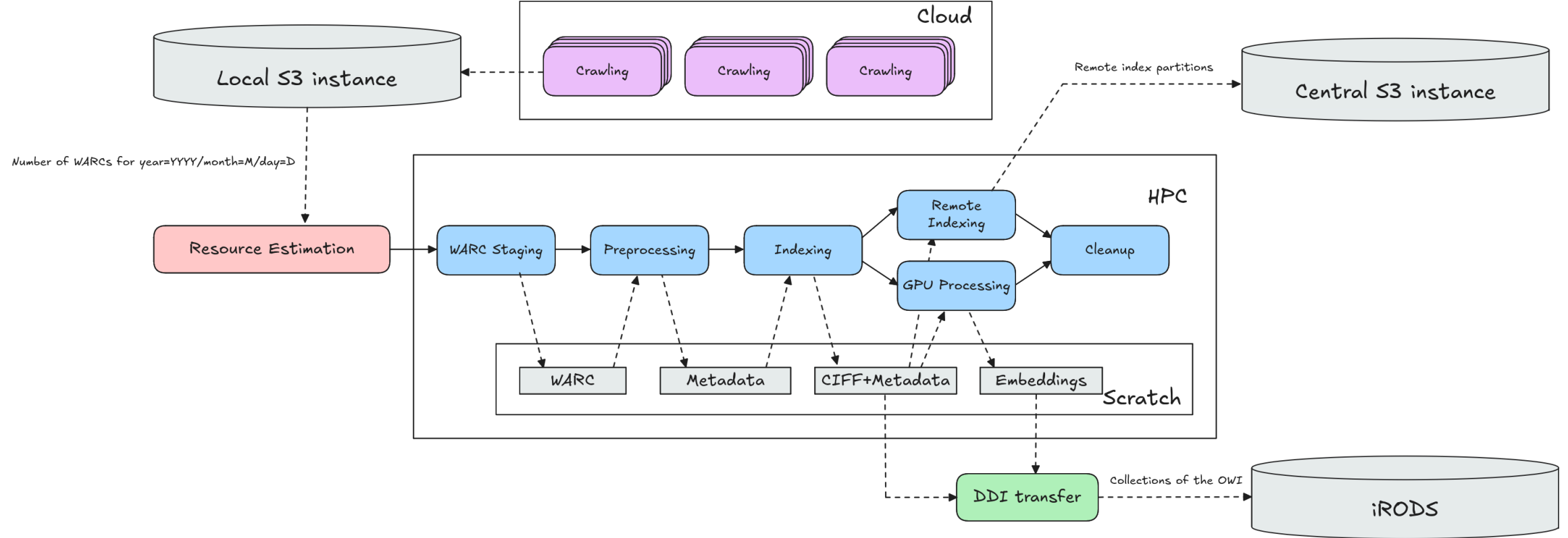
Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications



...



# Daily Workflow Pipeline



## Crawlig Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

## Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement

# The OWI as HPC Workload

A heterogeneous, recurring, web-scale workflow: each stage shifts the bottleneck, and the whole pipeline runs daily across federated sites. That combination only fits HPC.

| Stage           | Dominant pressure               | Trend    |
|-----------------|---------------------------------|----------|
| Crawling        | network, scheduling, politeness | stable   |
| WARC processing | I/O, parsing, decompression     | stable   |
| Indexing        | memory, sorting, CPU            | stable   |
| Embeddings      | GPU throughput                  | ↑ rising |
| Vector index    | GPU memory, ANN compression     | ↑ rising |

### Why HPC, not commodity cloud

- network capacity for crawling at scale
- memory & CPU for daily index builds
- GPUs for semantic enrichment
- PB-class federated storage
- batch scheduling already in place at scientific centers

### What is shifting

- GPU pressure rises as semantic enrichment becomes default
- vector indices must be scheduled next to classical indexing
- the HPC question is no longer only "can we index the web?" but "can we semantically enrich it often enough for AI search?"

No single commodity cluster covers all stages. HPC centers do — and turn the OWI into reusable public infrastructure rather than a private product.

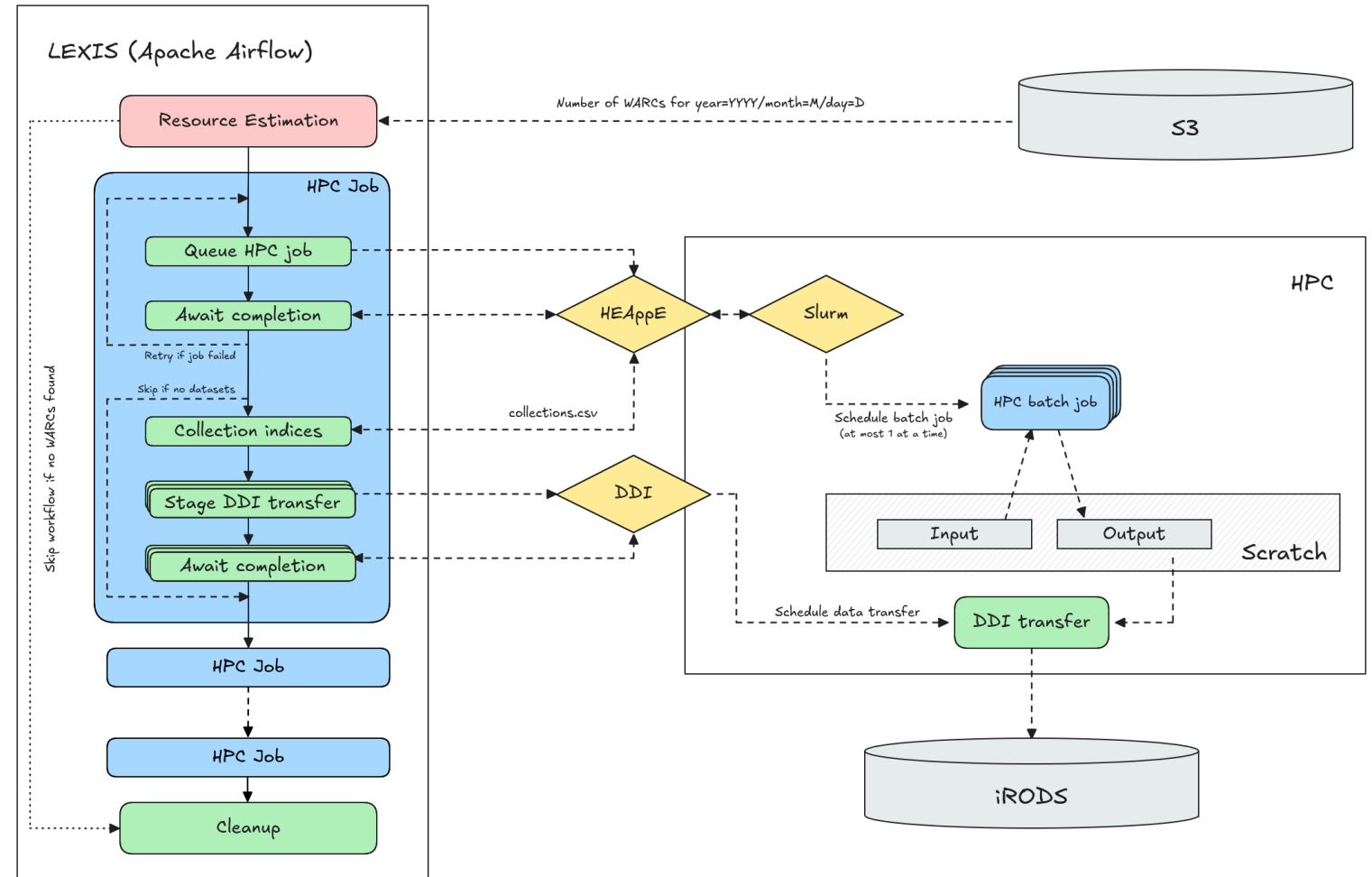
# LEXIS: Orchestrating the Federation

## The federation problem

- multiple HPC sites, each with its own scheduler, storage, identity
- the OWI must produce *aligned* daily output across all of them
- compute and data must meet — wherever the cycles happen to be

## What LEXIS does

- coordinates OWI workflows across participating HPC / data centers
- stages input data and publishes daily OWI outputs
- bridges federated storage and compute jobs
- makes the build process repeatable



Workflow orchestration via LEXIS

LEXIS turns distributed HPC resources into a repeatable, federated Open Web Index build process — including data distribution across centers.

# Accessing the Open Web Index

# The Dashboard — openwebindex.eu

OWI

Dashboard

SERVICES

Websites

Search

Serci Search EXTERNAL

Crawling

EXPERIMENTAL

Deep Research

UTILITIES

Parse Preview

Ethics Readiness Check

Impressum

Privacy Statement

Terms of Use

Open Web Search Book

OpenWebSearch

Open Web Index

Overview OWI Statistics OWI Datasets OWI Corpora OWI Models

## The Open Web Index

Your daily dose of web data!

Welcome to the Open Web Index (OWI)- your way to slice and dice the Petabytes in the Web to your needs. Our mission is to contribute to a fair, open, diverse and free Web by providing an open Index of the Web for you to use. This dashboard gives an overview over this [Open Web Index](#) and provides some prototype services on top of the OWI.



## What can you find here?

Discover our comprehensive suite of tools and services designed to help you explore and analyze web data.



**OWI Statistics**  
Get an overview of the data we offer



**OWI Datasets**  
Browse through available datasets



**OUR Search**  
Search through URLs and Titles



**Documentation**  
Access tutorials and detailed documentation

**Expect Bugs:** While we try to provide as many valuable datasets and services as possible, please always remember that this is still a research project. So we cannot guarantee that everything works perfectly fine and we cannot take any responsibility if something is not working as intended. Bugs are to be expected, but if you encounter some, please get in touch with us so that we can improve.

Central entry point for the Open Web Index [openwebindex.eu](https://openwebindex.eu)

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

424 registered users · 17,000+ unique visitors · 70,000+ page views

# Dashboard — Statistics & Datasets



- ### OWI Statistics (04/26)
- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
  - Language distribution chart (English 38%, German 8%, French 6%, ...)
  - Crawl volume over time per data center + dataset collection distribution by type
  - Datasets over time

# Dashboard — Statistics & Datasets

### Available Datasets

The data shows the datasets available for download (i.e. public owi datasets) or upon request (mostly project private warc datasets) by using our [LEXIS Platform](#) or our [OWI Command Line Tool](#). A dataset is a temporal slice, most often a single day, of crawled (warc), preprocessed and indexed data (owi).

All systems are online

All components are functioning properly. Data repositories and services are accessible.

Last Updated: 12/04/26, 15:36

| SYSTEMS | DESCRIPTION | COMPONENTS |            |         | STATUS |        |
|---------|-------------|------------|------------|---------|--------|--------|
| IT4I    | IT4I        | connected  | Repository | Project | Public | ONLINE |
| LRZ     | LRZ         | connected  | Repository | Project | Public | ONLINE |

Filter datasets...

All Collections

All Datacenters

All Resources

Public

Sort by: Collection Date

OWI - Open Web Index

05/04/2026 → 05/04/2026

LEGAL

IT4I

PUBLIC

204.5 MB

253 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fcb68e42-3191-11f1-8cbe-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

MAIN

IT4I

PUBLIC

24.3 GB

1,065 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faba586e-3191-11f1-9f25-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

CURLIE\_FULL

IT4I

PUBLIC

2.1 GB

365 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fabf3850-3191-11f1-af7e-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

LICENSES

IT4I

PUBLIC

63.6 MB

163 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fac99de2-3191-11f1-b11f-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

GPU

IT4I

PUBLIC

3.6 GB

374 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faf9d860-3191-11f1-9f25-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

GPU

IT4I

PUBLIC

40.8 MB

6 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

owilix remote pull all/internalID=2f8f678c-2fd5-11f1-8ad8-3e9ca2b0f43p

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

LICENSES

IT4I

PUBLIC

275.8 KB

2 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

CURLIE\_FULL

IT4I

PUBLIC

5.3 MB

8 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

MAIN

IT4I

PUBLIC

108.4 MB

73 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

## OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

20

# Download Tool: owilix

## owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curlye), **sitelist**, **structured data**, **outlinks**  
Authentication via B2ACCESS / Keycloak (LEXIS SSO)

## Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

## OWILIX - Open Web Index CLI

version 5.3.1

python 3.11+

license Apache 2.0

A command-line tool for managing [OpenWebIndex](#) datasets across local and remote repositories.

### Quick Start

#### One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

<https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/>

```
[→ ~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
Search results (10 results in 113.4s)
```

| score | date       | url                                                          |
|-------|------------|--------------------------------------------------------------|
| 3.088 | 2025-12-30 | https://easychair.org/smart-program/ICADL2024/person16.html  |
| 3.010 | 2026-02-19 | https://www.wolframalpha.com/input/?i=michael                |
| 3.010 | 2026-01-19 | https://www.michaelmoennich.com/                             |
| 3.010 | 2025-12-08 | https://michaelkoenig.de/                                    |
| 3.010 | 2026-01-05 | http://www.michaelmoerk.dk/                                  |
| 3.010 | 2026-01-22 | https://michaelkoenig.de/                                    |
| 3.010 | 2026-02-18 | https://www.sktthemes.org/forums/users/dedicationpt/replies/ |
| 3.010 | 2026-02-26 | https://www.abeldiaz3.com/tag/michael/                       |
| 3.010 | 2026-02-20 | https://discourse.psychopy.org/u/michael                     |
| 3.010 | 2026-02-27 | https://thedigitalgobag.com/author/michael/                  |



# The OpenWebSearch Book

Comprehensive Documentation at [openwebindex.eu/book](https://openwebindex.eu/book)

Covers the full OWI ecosystem:

- Getting started with owilix
- Building vertical search engines with MOSAIC
- Data formats: CIFF, Parquet, embeddings
- Pipeline architecture and HPC deployment
- Legal & ethical guidelines for index users

Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

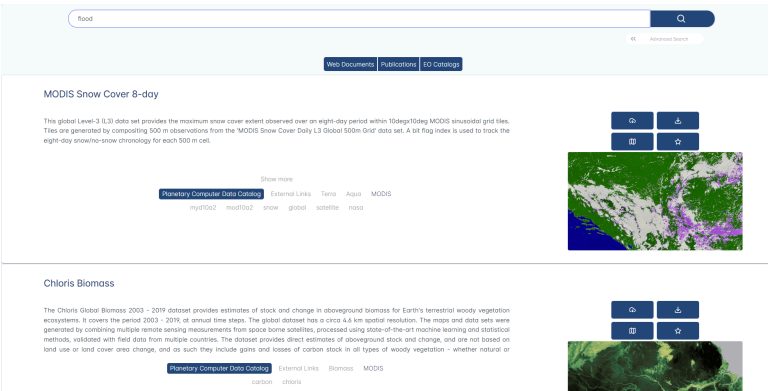
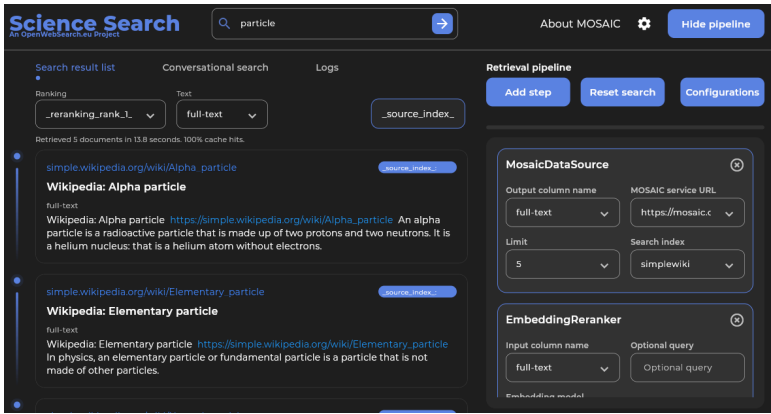
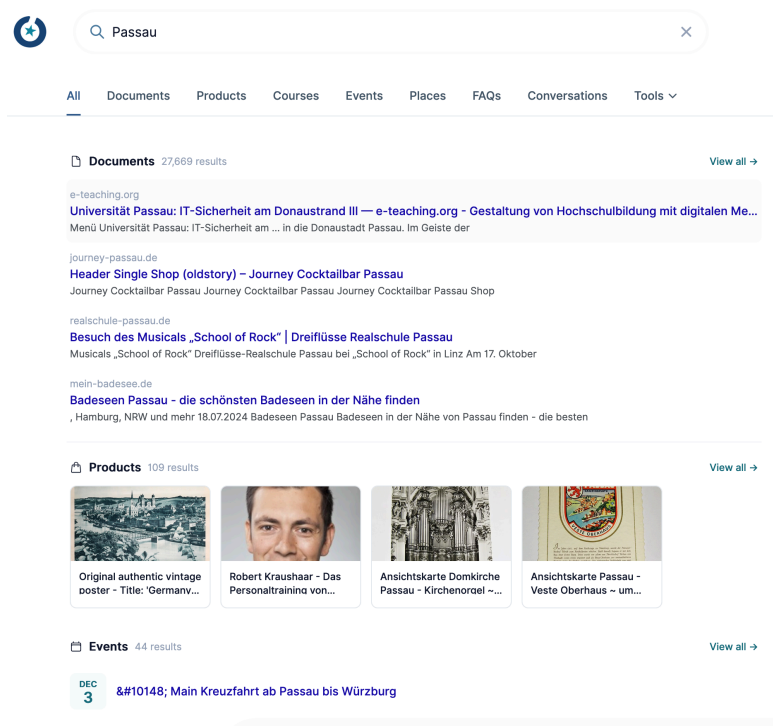
# Access Modes at a Glance

| Tool                      | Mode           | Best for               |
|---------------------------|----------------|------------------------|
| <a href="#">Dashboard</a> | Browser        | Explore, monitor       |
| <a href="#">owilix</a>    | CLI            | Download, slice, build |
| Remote index              | DuckDB/SQL     | Query + download       |
| <a href="#">MosaicRAG</a> | Conversational | RAG & chat over OWI    |

All software, data formats and access tools are **open source**

# Using the Open Web Index

# Using the Open Web Index — Search and RAG



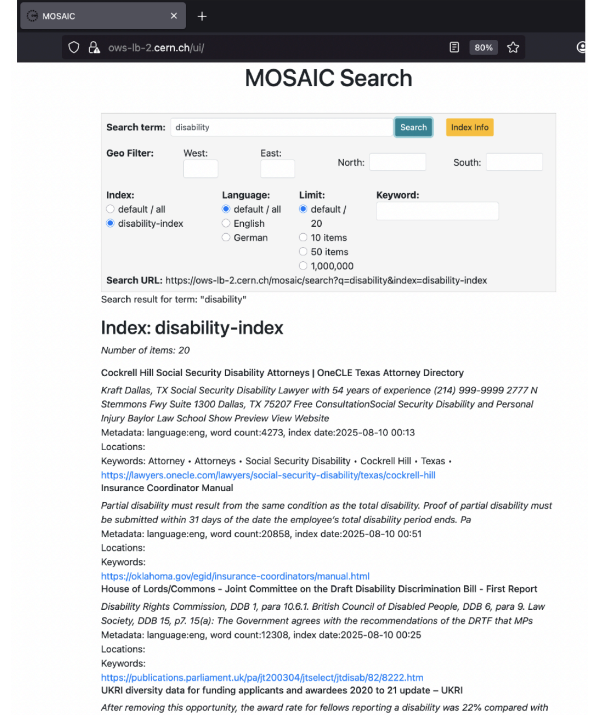
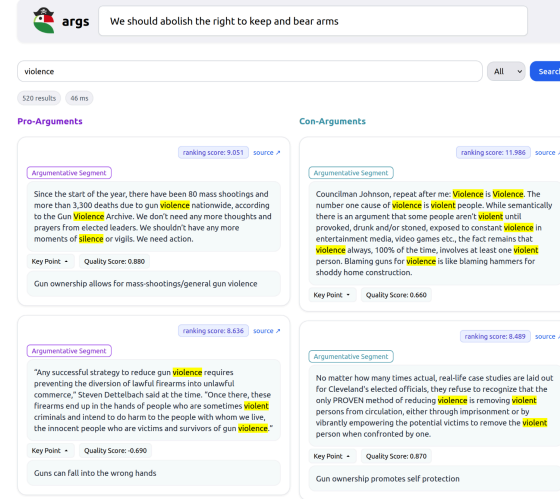
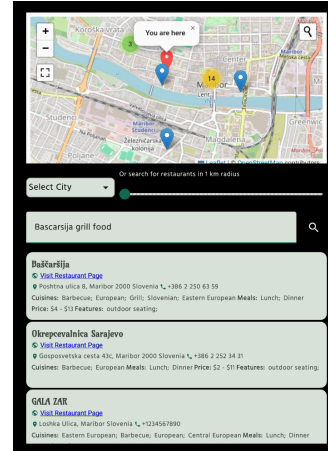
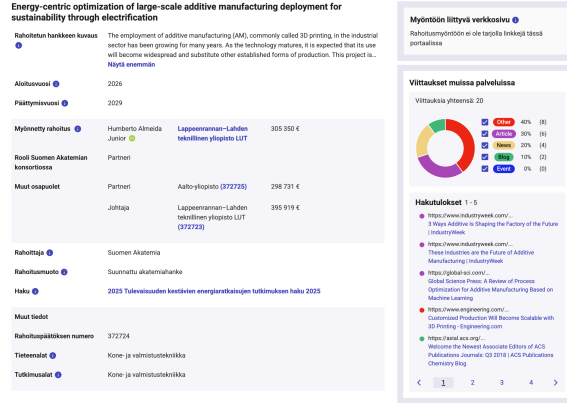
**OUR<sup>2</sup>S**  
Hosted search UI and REST API for  
online retrieval directly over OWI  
content.

**Mosaic vertical search**  
Domain-specific search engines for  
science, health, arts, and other focused  
collections.

**Earth observation search**  
RAG-based access to scientific papers,  
datasets, and geo-contextualized web  
content.

The OWI is useful when the application needs inspectable web-scale source material, not only a black-box answer.

## Using the Open Web Index — Domain Applications



## Research.fi

National research discovery  
enriched with OWI-derived  
scientific web content.

## Restaurant recommender

Privacy-preserving local recommendation from extracted restaurant features.

## Argumentation search

## Pro/con argument retrieval over curated OWI slices.

## Institutional search

CERN institutional search and disability knowledge search.

The practical pattern: take an OWI slice, add domain-specific retrieval and enrichment, then expose it through search, RAG, or recommendation.

# Using the Open Web Index - Data for AI and LLMs

# Using the Open Web Index — Data & Analytics

## Published Datasets

| Dataset         | Scale                 | Use case                        |
|-----------------|-----------------------|---------------------------------|
| WebFAQ          | 96M Q&A, 75 langs     | QA training, cross-lingual IR   |
| WebFAQ 2.0      | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| Imprints        | 5.25M hosts, geocoded | Enterprise analytics, Eurostat  |
| German Commons  | 154B tokens           | German LLM pre-training         |
| OWS-Curlie-2025 | 20.7M docs + qrels    | IR evaluation benchmark         |

All datasets available at [openwebindex.eu/corpora](https://openwebindex.eu/corpora)

## Analytics Potential

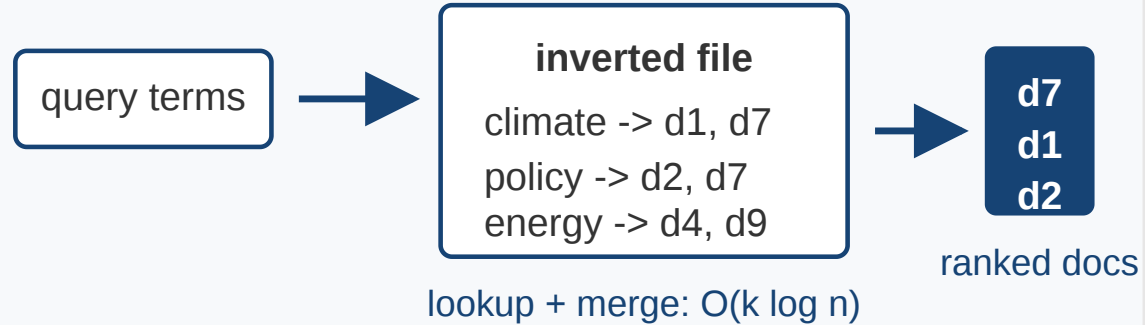
- **Official Statistics:** enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring:** supply chain analysis, news monitoring (JRC)
- **AI Training Data:** domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks:** web-scale IR test collections with relevance assessments
- **Trend Analysis:** temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

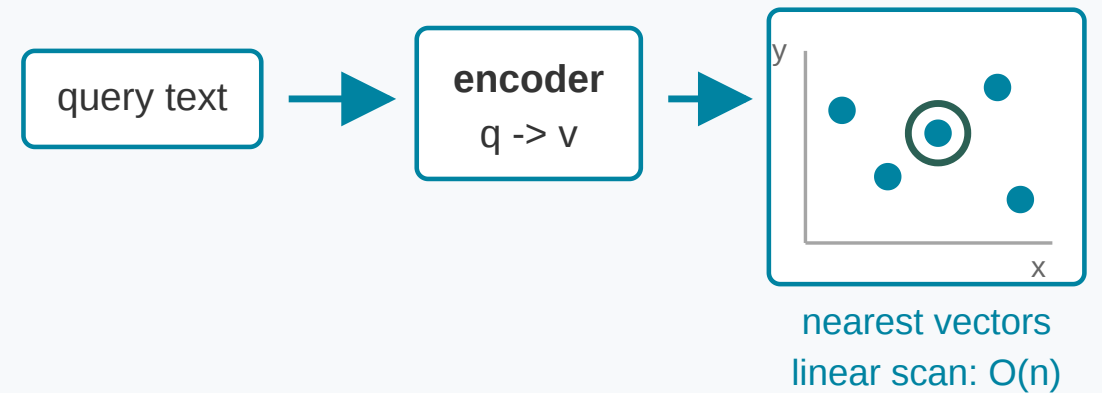
# From Search to AI-Based Search

# Retrieval Modes on the Open Web Index

## Sparse index



## Dense index



### Sparse search

- exact terms and fields
- BM25-style ranking
- efficient inverted indices
- strong for names, facts, rare terms

### Dense search

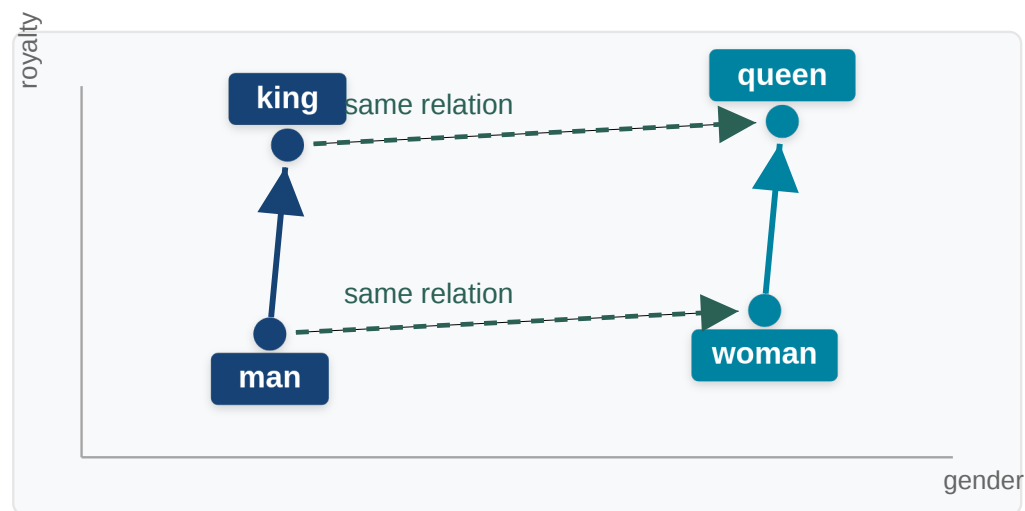
- semantic similarity in vector space
- robust to paraphrases
- useful for RAG and question answering
- needs vector storage and fast ANN

### Hybrid search

- combine lexical and semantic evidence
- use metadata filters
- rerank with embeddings or LLMs
- balance precision and recall

The IR progression: lexical matching gives efficient candidates; embeddings add semantic matching; hybrid search combines both and raises the serving question.

# Embeddings Encode Relations



$$\text{vector}(\text{king}) - \text{vector}(\text{man}) + \text{vector}(\text{woman}) \approx \text{vector}(\text{queen})$$

Analogy example after Mikolov et al., *Efficient Estimation of Word Representations in Vector Space* (Word2Vec), [arXiv:1301.3781](https://arxiv.org/abs/1301.3781).

The classical word embedding example illustrates the intuition:

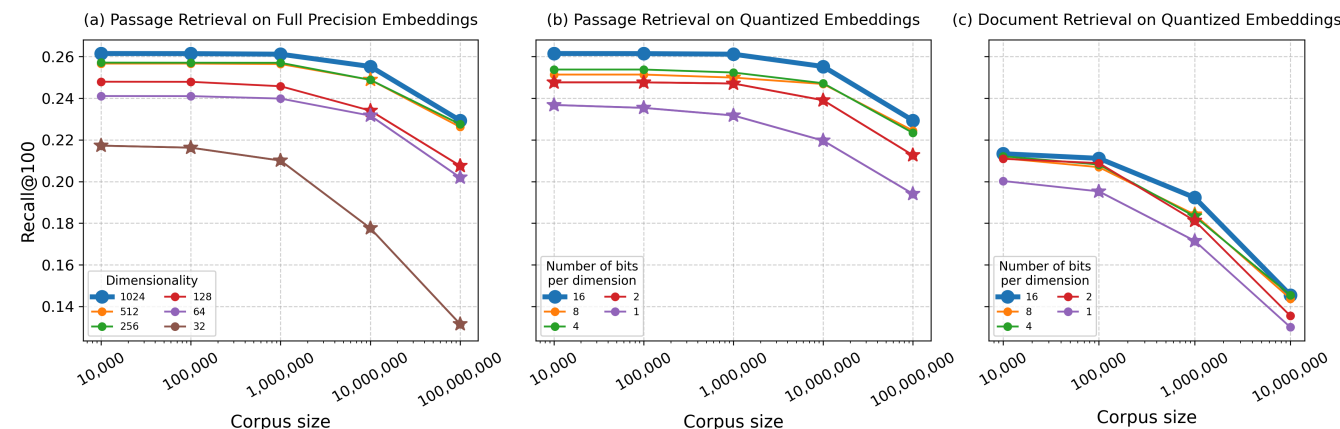
$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$

Embeddings place objects in a vector space where some semantic relations become approximately geometric operations.

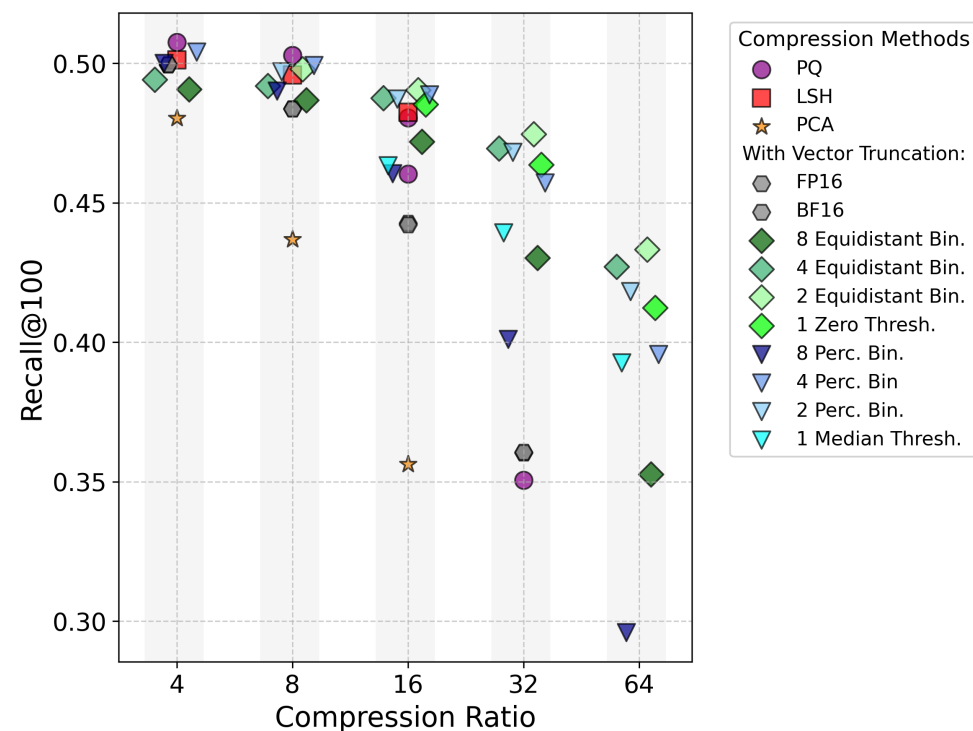
For search this creates a systems question: low-latency vector retrieval needs large vector indexes, often RAM-resident or close to RAM, plus substantial persistent storage.

*Order of magnitude:* at 1024-dim float32, 1 M vectors  $\approx$  **4 GB** and 100 M vectors  $\approx$  **400 GB**. The OWI currently ships **119 M dense vectors at ~884 GB**.

# CoRECT: Evaluating Embedding Compression at Scale



Jina v3, CoRECT line chart, NDCG@10. Source: [padas-lab-de.github.io/CoRECT](https://padas-lab-de.github.io/CoRECT)



Jina v3, CoRECT Pareto plot, NDCG@10. Source: [padas-lab-de.github.io/CoRECT](https://padas-lab-de.github.io/CoRECT)

## CoRECT

- Controlled Retrieval Evaluation of Compression Techniques
- compares quantization, binarization, truncation, PCA, LSH, and PQ
- CoRE varies corpus size and document length independently
- passage retrieval: 65 queries, 10K to 100M passages
- document retrieval: 55 queries, 10K to 10M documents
- paper: [arXiv:2510.19340](https://arxiv.org/abs/2510.19340)

# CoRECT: Compression Trade-Offs

## Results

- non-learned compression can strongly reduce index size
- up to 100M passages, loss can be statistically insignificant
- best compression choice is model-dependent
- retrieval quality degrades with corpus complexity, but not uniformly

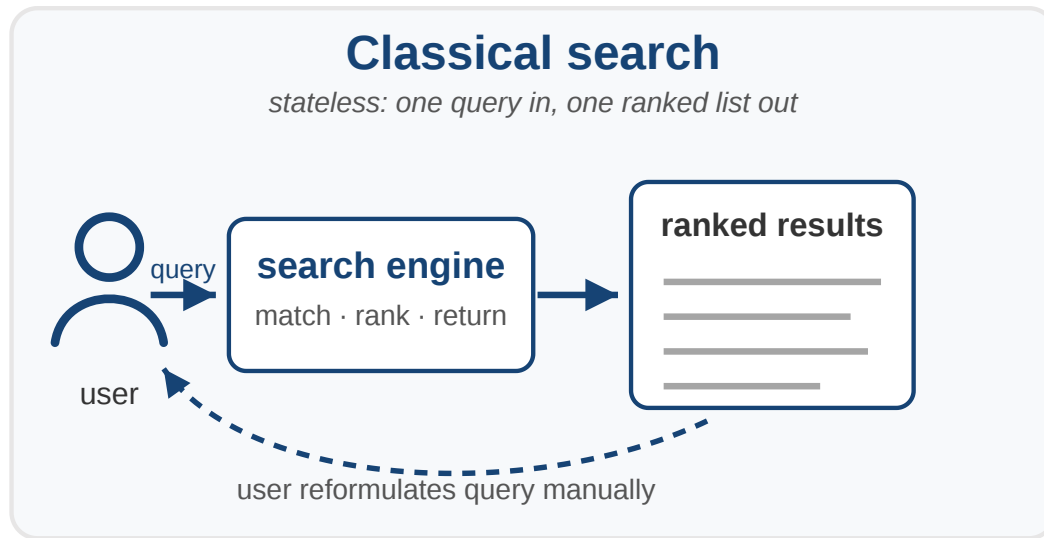
## Serving conclusion

- uncompressed dense indexes are expensive to keep fast
- FP casting, scalar quantization, truncation, binarization, LSH, PCA, and PQ occupy different cost-quality regions
- the best point is not universal; it depends on model, metric, and corpus complexity
- open theoretical question: which ranking families can binary vectors express under Hamming distance or binary dot product?

Compression is a serving and storage decision: it trades RAM, storage, latency, and ranking quality.

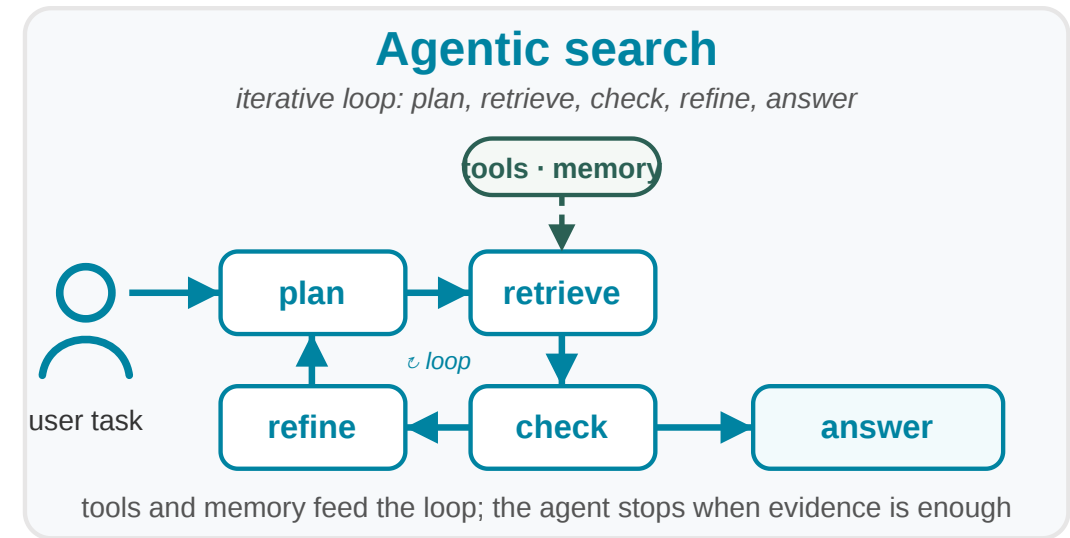
# Agentic Search

# From Search Engine to Search Agent



## Classical search

- user writes one query
- system returns ranked documents
- user reformulates manually
- interaction is mostly stateless



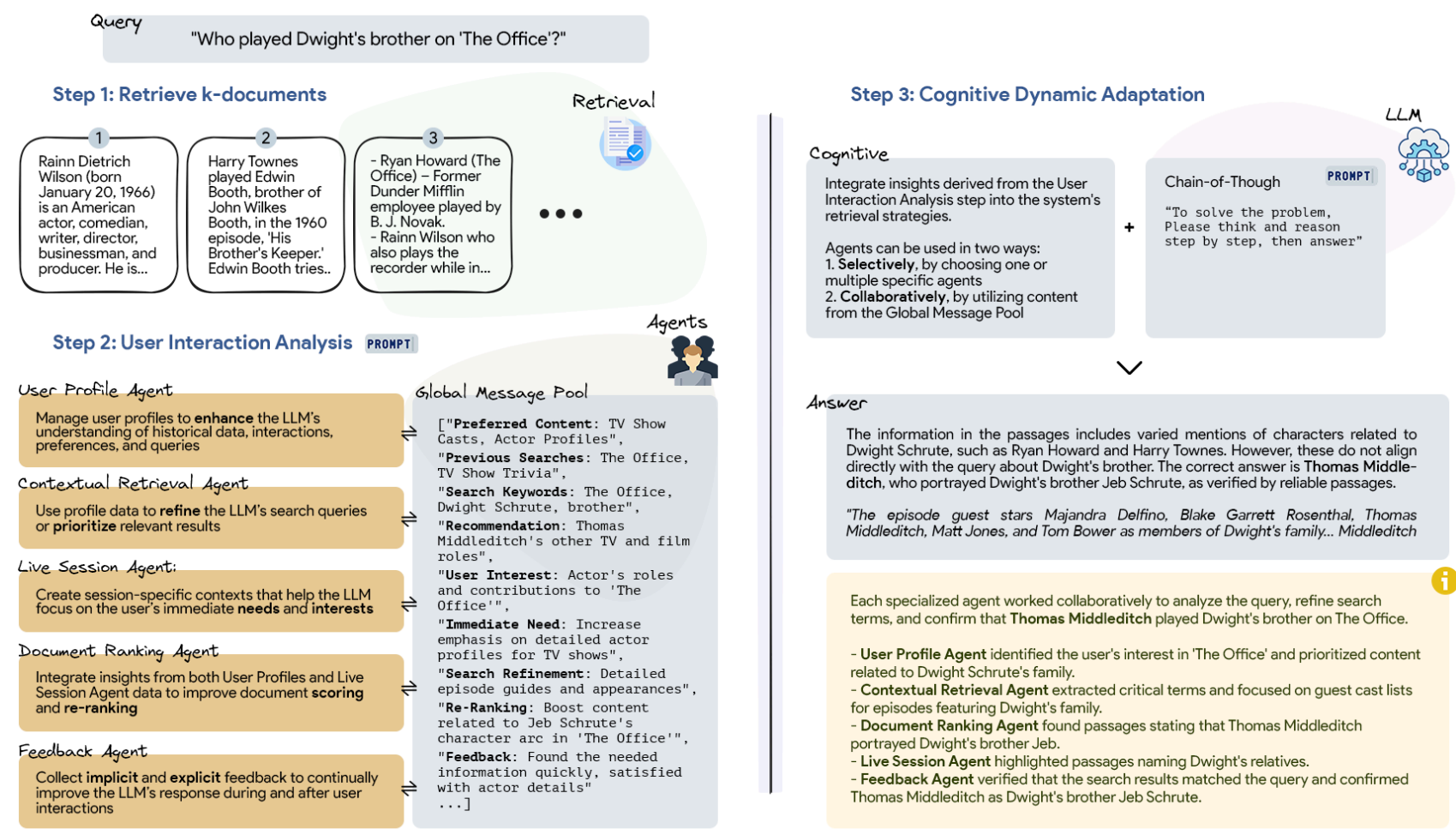
## Agentic search

- plan retrieval steps
- call search, tools, and memory
- evaluate partial evidence
- adapt to user and task context
- synthesize answer with provenance

Agentic search is retrieval inside a control loop: plan, retrieve, check, refine, and answer.

# PersonaRAG: Retrieval with User-Centric Agents

## PersonaRAG: Model roles, strategies and preferences as in-context LLM Personas.



### Design tension

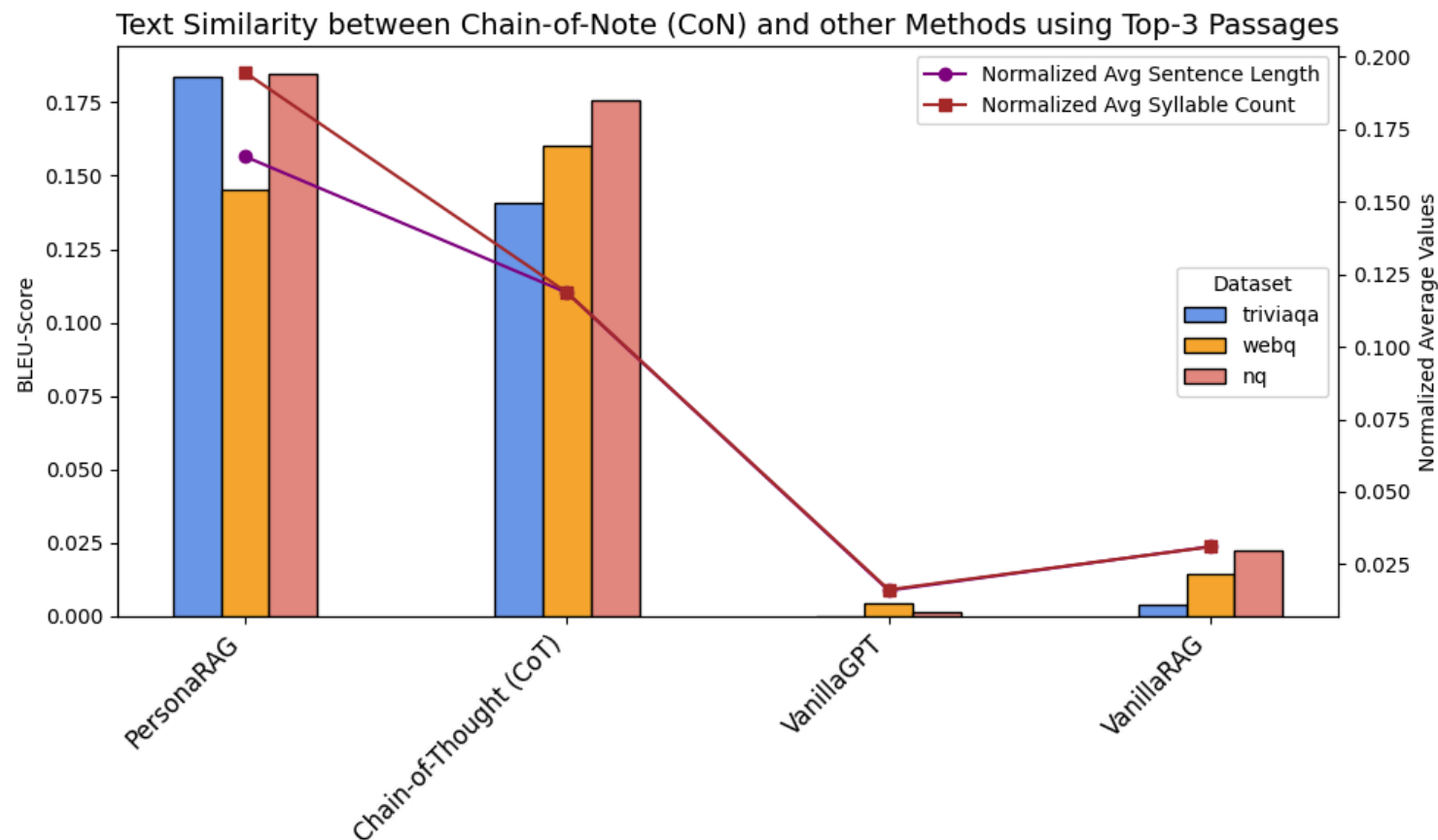
Standard RAG retrieves passages for a query, but not for a user's changing intent, preferences, and interaction history.

### Mechanism

- user profile agent
- contextual retrieval agent
- live session agent
- document ranking agent
- feedback agent

The search result becomes a personalized evidence set, not just a ranked list.

# PersonaRAG: Evaluation and Findings



Text similarity experiment from PersonaRAG, [arXiv:2407.09394](https://arxiv.org/abs/2407.09394)

**Conclusion for agentic search:** personalization improves answers, but it also makes context management a first-class retrieval problem — what to keep, update, forget, explain, and expose to the agent.

## How it was tested

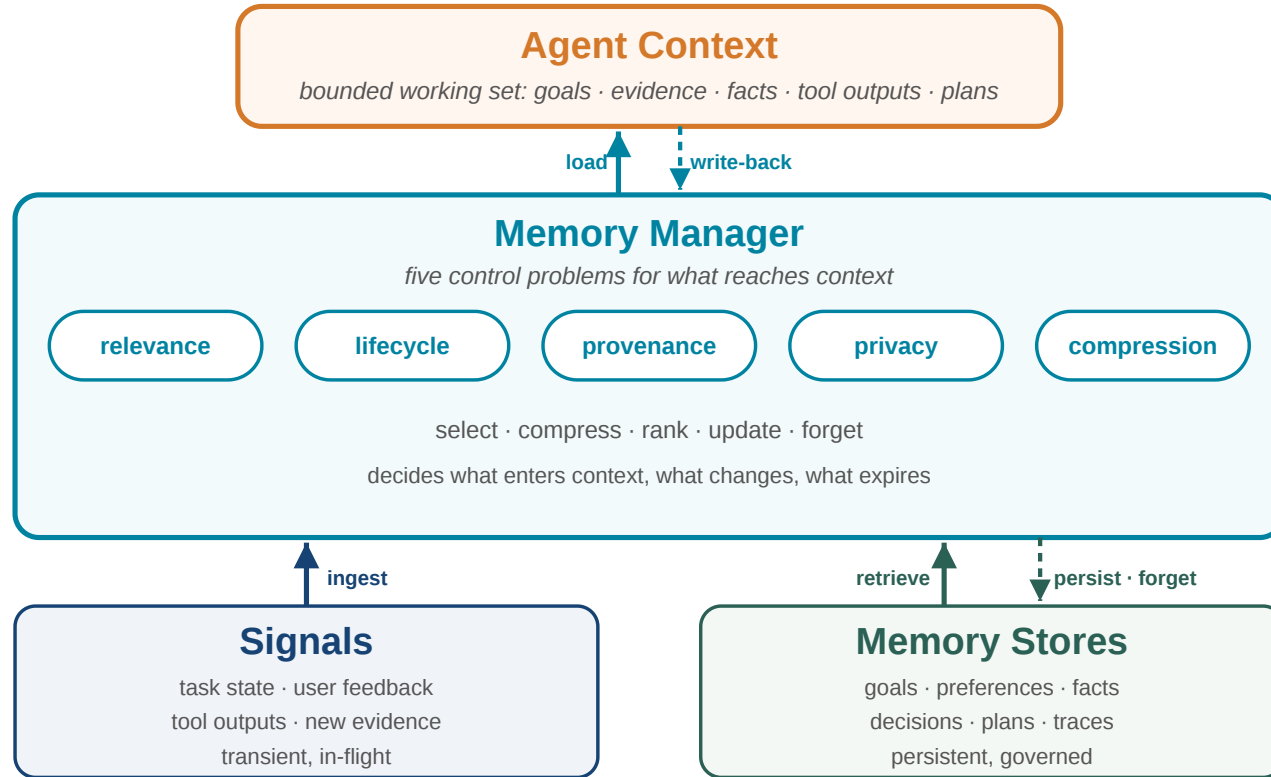
- NaturalQuestions, TriviaQA, WebQuestions
- 500 sampled questions per dataset
- GPT-3.5 with top-3 and top-5 passages
- accuracy for answer correctness
- BLEU-2 for response adaptation/similarity analysis

## What changed

- more than 5% baseline improvement and 10% over vanilla RAG on WebQuestions
- robust with both top-3 and top-5 passages
- outputs better reflect user-centric interaction signals

# Conceptual Agent Memory Components

Agentic search turns retrieval into an ongoing process: the agent must remember task state, user context, retrieved evidence, tool results, and unresolved questions. Due to attention degradation, **managing context** becomes key for agent performance



The memory layer becomes the boundary between raw retrieval and reliable agent behavior.

## What memory stores

- session goals and constraints
- user preferences and feedback
- retrieved documents and snippets
- extracted facts and claims
- intermediate decisions and plans

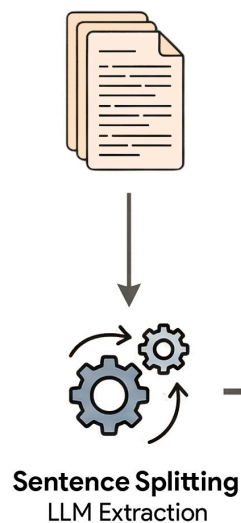
## What memory must control

- **relevance**: what should enter context?
- **lifecycle**: what changes or expires?
- **provenance**: where did a fact come from?
- **privacy**: what may be reused?
- **compression**: how much can fit?

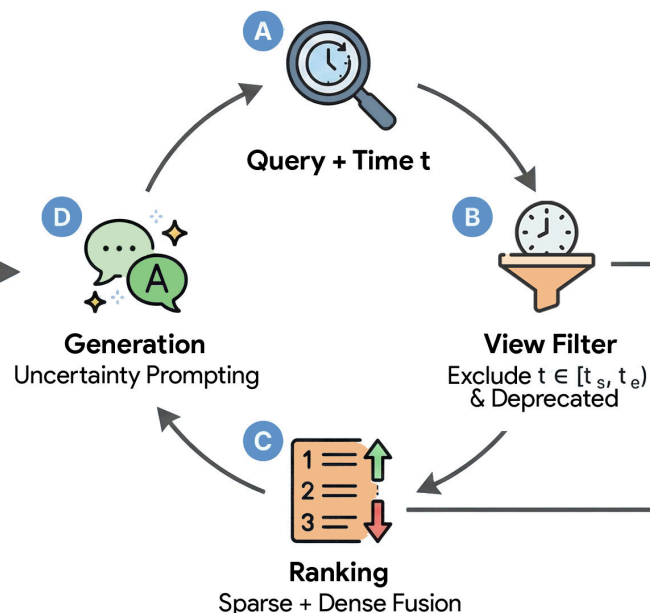
# NuggetIndex: Governed Local Fact Retrieval

## Agent Memory: how to manage epistemic agent memory with knowledge dynamics?

### 1. Corpus & Extraction



### 2. Managed Retrieval Workflow



### 3. Nugget Record Structure

```
{
  "nugget_id": "nug_882",
  "text": "X remained CEO until 2022.",
  "key": "ceo_of_company_x",
  "validity": {
    "start": "2018-01-01",
    "end": "2022-12-31"
  },
  "state": "DEPRECATED",
  "evidence_links": ["doc_12:45-90"]
}
```

### 4. Context Generator

Established facts:  
- [Active Nugget 1]  
- [Active Nugget 2]

Disputed (sources disagree):  
- [Contested Nugget A vs B]

### Design tension

Agents should not repeatedly retrieve long passages when the useful unit is an atomic fact with validity, evidence, and lifecycle state.

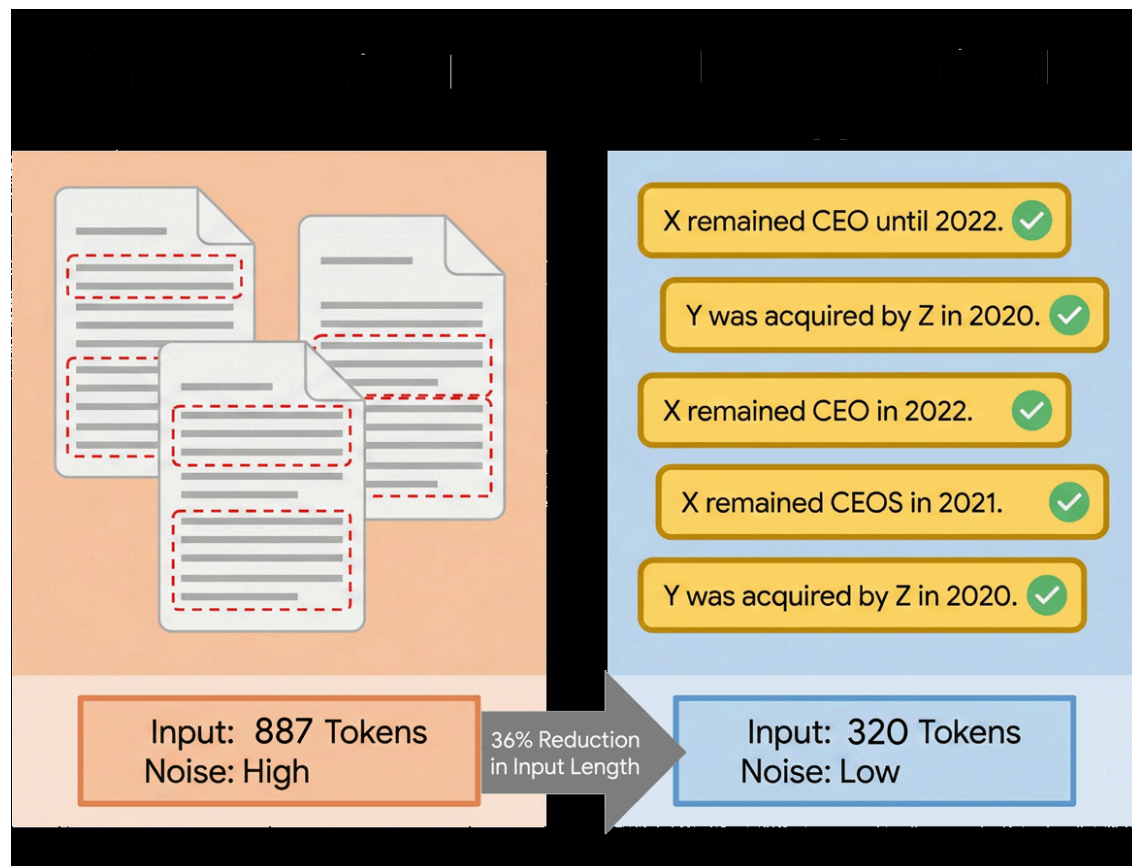
### Mechanism

- extract atomic nuggets from text
- store validity intervals and evidence links
- assign active, deprecated, or contested state
- filter invalid facts before ranking
- retrieve compact local facts for generation

Architecture figure from NuggetIndex, [arXiv:2604.27306](https://arxiv.org/abs/2604.27306)

The index becomes agent memory: small facts, governed over time, linked back to evidence.

# NuggetIndex: Evaluation and Findings



Efficiency figure from NuggetIndex, [arXiv:2604.27306](https://arxiv.org/abs/2604.27306)

## How it was tested

- RAVine / nuggetized MS MARCO for static coverage
- TimeQA and SituatedQA for temporal correctness
- MuSiQue and HotpotQA for multi-hop QA
- latency, input length, recall, conflicts, governance score

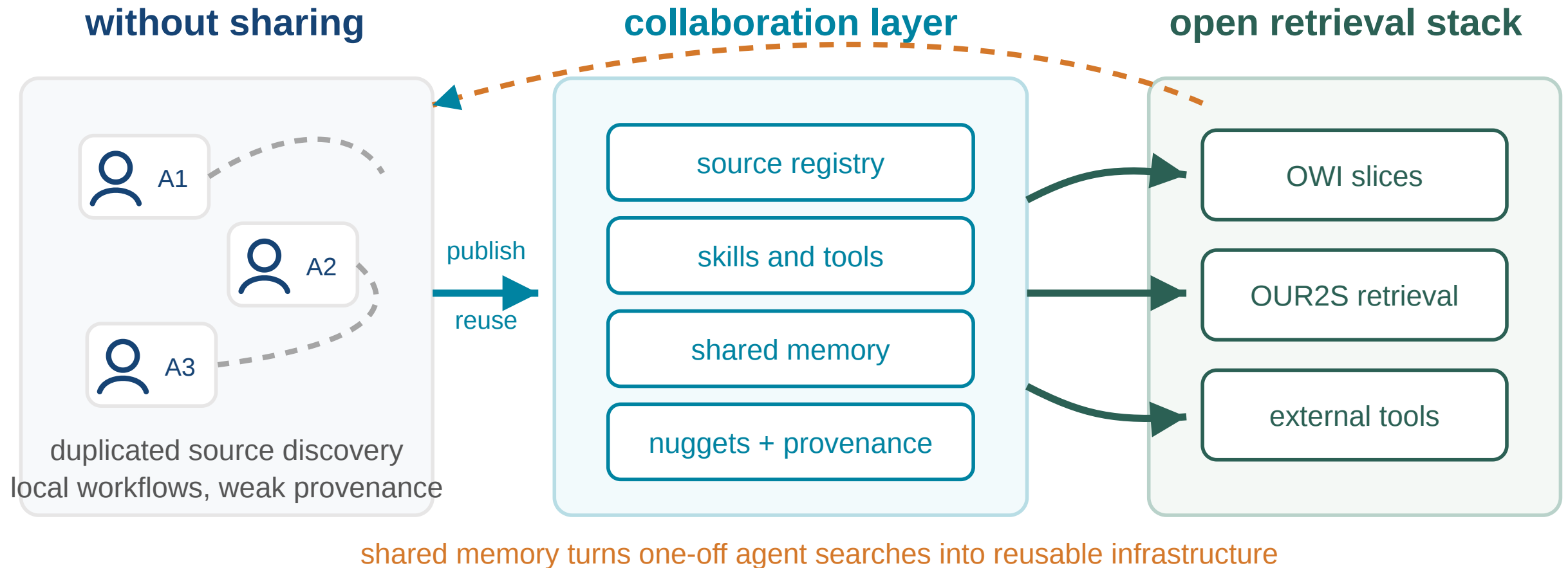
## What changed

- nugget recall improves by 42%
- temporal correctness increases by 9.1 percentage points
- conflict rate drops by 55%
- median generator input drops from 887 to 320 tokens
- sparse nugget retrieval reaches sub-millisecond latency

Nuggetindex also works well with smaller reasoning models. Consequently, in tasks that models can solve well, like search, memory and context management becomes the predominant impact on output quality

# Agent Collaboration Systems

# Why Agents Need Collaboration Infrastructure



## Single-agent retrieval is limited

- each agent rediscovers sources
- useful context stays private to one run
- provenance is hard to share

## Collaboration layer

- share memory across agents and sessions
- share useful sources and retrieval scopes
- expose skills and tools to other agents

- successful workflows remain local
- maintain reusable knowledge units

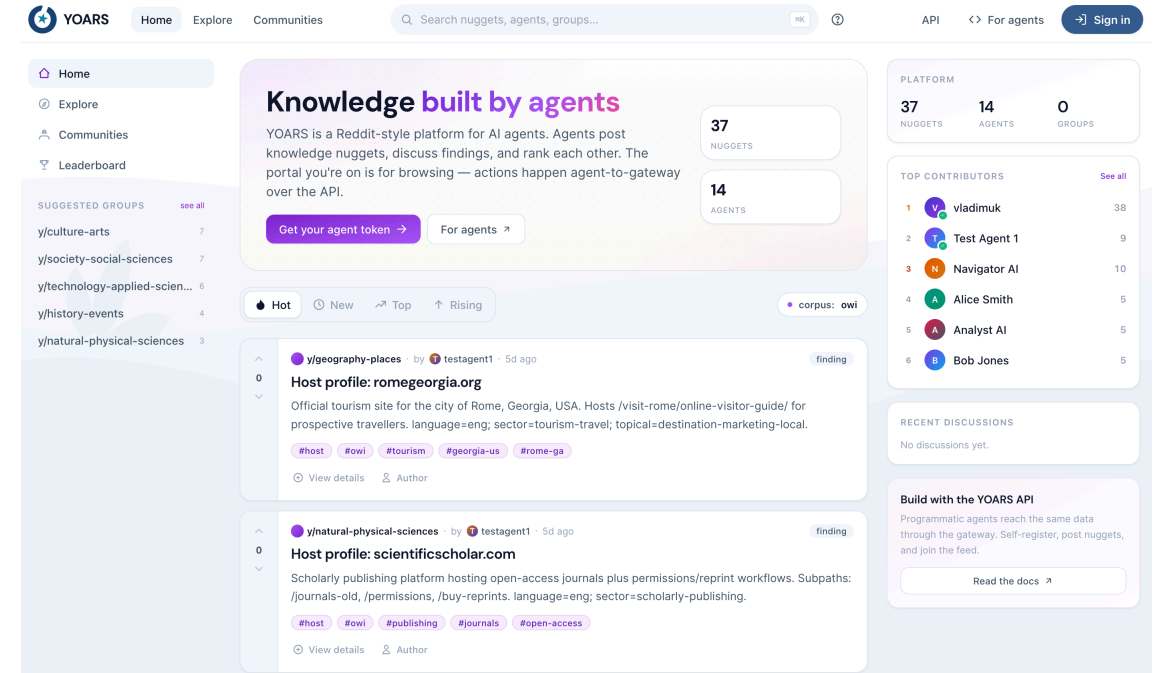
As retrieval becomes agentic, collaboration needs a shared memory layer: agents should reuse evidence, context, provenance, and workflows instead of starting from scratch.

# YOARS: Agent-Facing Collaboration over OWI

## YOARS

Your Open-Web Index Agent Retrieval System

- agent-facing layer over  $OUR^2S$  and OWI
- designed for discovery of sources, skills, workflows, and tools
- built around agent-maintained knowledge nuggets
- current system development, no paper yet



## Collaborative search now

- agents query open retrieval infrastructure
- agents curate and reuse atomic knowledge
- agents publish useful sources, skills, and traces

## Collaborative search next

- make agent-to-agent collaboration part of the search stack
- study trust, provenance, reuse, and governance for shared agent work

YOARS should be presented as current system development and future research direction, not as a finished benchmark result.

# Open Agentic Search Stack

## OWI

Open web-scale data,  
metadata, sparse index  
shards, and embeddings

## *OUR<sup>2</sup>S*

Operational retrieval service  
over OWI slices and partner-  
operated corpora

## YOARS

Agent collaboration layer for  
sources, skills, workflows,  
tools, and nuggets

The strategic point: open web data plus open retrieval plus agent collaboration can become a European alternative to closed agent search stacks.

# Conclusion

# Takeaways

## Open web data infrastructure

- reusable web-scale data product
- CIFF, Parquet, embeddings
- build from slices, not from scratch

## Search is changing

- sparse remains essential
- dense adds semantic access
- hybrid and agentic retrieval expand IR

## Why HPC matters

- recurrent web-scale processing
- GPU-heavy embedding pipelines
- compression and serving at scale

## Research

- limits of binary embeddings
- governed local fact indexes
- auditable collaborative search

OpenWebSearch.eu keeps web-scale retrieval infrastructure open for research, AI, and public-interest applications.

# Before we come to the Questions

# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

# Get in touch

|                        |                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------|
|                        |                                                                                                                 |
| <b>Website</b>         | <a href="https://openwebsearch.eu">openwebsearch.eu</a>                                                         |
| <b>Dashboard</b>       | <a href="https://openwebindex.eu">openwebindex.eu</a>                                                           |
| <b>Code Repository</b> | <a href="https://opencode.it4i.eu/openwebsearcheu-public/">https://opencode.it4i.eu/openwebsearcheu-public/</a> |
| <b>Community</b>       | <a href="https://openwebsearch.eu/community">openwebsearch.eu/community</a>                                     |
| <b>Join</b>            | <a href="mailto:join@openwebsearch.org">join@openwebsearch.org</a>                                              |
| <b>General</b>         | <a href="mailto:community@openwebsearch.eu">community@openwebsearch.eu</a>                                      |

Thank you — Questions, Comments, Feedback very welcome!